1.

a. The attention mechanism allows the model to dynamically focus on which part of the sentence is most important to the current word being generated. Not all tokens should be used with equal relevance to a given output position because they aren't equally relevant. For example, the word Please isn't nearly as relevant as the word attention when the model is trying to translate the idea of attention. How the model determines the relevant tokens to the current translation is by learning to assign attention weights, a probability distribution over all source tokens in relation to the current output position. It queries when it needs an output to the available keys, and then takes a weighted average of the actual content to inform its next output.

b. The query asks for what tokens are relevant to the concept at hand. Projecting the input into a dedicated query space instead of using raw embeddings is preferable because it allows for optimization for search and matching against keys, where the raw embedding would be less effective.

c. The key represents an answer to the question the query is asking. The dot product, then, is maximized when the key matches the query most effectively.

d. The value is the actual content of the token and its meaning that gets utilized. The key and value are different in that the key is optimized for retrieval and matching, whereas the value is the useful content that flows forward in the network.

e. They would be different, because they hold different uses and roles in the sentence. The atensan word is used to reference the technical subject while the

actual word for attention and awareness, sraddha, holds entirely different content and usage.

Problem 2

a) $S = K^T Q \rightarrow K^T = \begin{pmatrix} 1 & 0 \\ 1 & 1 \end{pmatrix}$

$$\begin{pmatrix} 1 & 0 \\ 1 & 1 \end{pmatrix}\begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} = \begin{pmatrix} 1 & 0 \\ 1 & 1 \end{pmatrix}$$

$d_s = 2$ $(i,j)$ is how much key $i$ matches query $j$

b) $\frac{S}{\sqrt{2}} = \begin{pmatrix} \frac{1}{\sqrt{2}} & 0 \\ \frac{1}{\sqrt{2}} & \frac{1}{\sqrt{2}} \end{pmatrix} = \hat{S}$

c) $\begin{pmatrix} 0.5 & \frac{1}{1+e^{\frac{1}{\sqrt{2}}}} \\ 0.5 & \frac{e^{\frac{1}{\sqrt{2}}}}{1+e^{\frac{1}{\sqrt{2}}}} \end{pmatrix}$ col 2: $e^0 + e^{\frac{1}{\sqrt{2}}}$ denominator

d) $Z = VA \rightarrow \begin{pmatrix} 2 & 0 \\ 0 & 3 \end{pmatrix}\begin{pmatrix} 0.5 & 0.3322 \\ 0.5 & 0.6678 \end{pmatrix}$

$$= \begin{pmatrix} 1.0 & 0.6644 \\ 1.5 & 2.0034 \end{pmatrix}$$

dimension $= 2$ or $2 \times 2$

e) The first column of A attends to both tokens equally, because their values both before and after softmaxing are identical and thus identified with both tokens equally.

Problem 3:

a) $$\text{softmax}(s) = \begin{pmatrix} \frac{e^2}{e^2+e+1} \\ \frac{e}{e^2+e+1} \\ \frac{1}{e^2+e+1} \end{pmatrix} = \begin{pmatrix} 0.6652 \\ 0.2447 \\ 0.09 \end{pmatrix}$$

$$0.6652 + 0.2447 + 0.09 = 1$$

b) $$\text{softmax}(s) = \begin{pmatrix} \frac{e^4}{e^4+e^2+1} \\ \frac{e^2}{e^4+e^2+1} \\ \frac{1}{e^4+e^2+1} \end{pmatrix} = \begin{pmatrix} 0.8438 \\ 0.1142 \\ 0.042 \end{pmatrix}$$

The distribution became much more peaked, with the max value dominating much more.

c) Large raw scores dominate much more in the gradient flow during training, and other scores disappear.

Problem 4

a) $X$: 6 tokens, 512 dimension model

$\Rightarrow 512 \times 6$

$W^Q$: dimensions of $W^Q = d_{model} \times d_k$

$\Rightarrow 512 \times 6$

$Q: (W^Q)^T X = (64 \times 512)(512 \times 6) = 64 \times 6$

b) $K$ has same shape as $Q$: $64 \times 6$

$K^T \Rightarrow 6 \times 64$

$K^T Q \Rightarrow (6 \times 64)(64 \times 6) = \boxed{6 \times 6}$

c) softmax doesn't change shape, still $6 \times 6$

d) $V \Rightarrow d_v \times n \Rightarrow 64 \times 6$

$VA = (64 \times 6)(6 \times 6) = \boxed{64 \times 6}$

Problem 5

a) Each of 3 matrices = $256 \times 64$

$\rightarrow$ 16384 parameters

single head : $3 \cdot 16384 = 49152$

4 heads : $4 \cdot 49152 = 196608$

b) Total for 1 layer: $W^o : (4 \times 64) \times 256$

$\rightarrow 256 \times 256 = 65536$

Total : $196608 + 65536 = 262,144$

c) Across all 6 layers: $6 \times 262144$

$= 1572864$

d) Factor change when dmodel doubles

$\rightarrow$ per head : $512 \times 128 = 65536$

$\times 4$ heads : $4 \times 3 \times 65536 = 786432$

$W^o = 512^2 = 262144$

Total per layer. $786432 + 262144 = 1048576$

Factor : $\frac{1048576}{262144} = \boxed{4}$

# Problem 6

1 Multi headed attention is preferred over single head attention because it allows for encoding of multiple different types of context. Single headed attention allows for distribution of context, to be encoded into the query, key, and value matrices, and loses out on competing information because of the single subspace. Multi headed attention, however, allows for multiple specializations to be concatenated into a vector, one for each head. Multi head essentially just allows for more complex understanding of context because it opens the door to multiple independent subspace representations of meaning.

2

i The model isn't using its full representational capacity. 6 of the heads are outputting information that could be captured by just one. It's not necessarily a problem, it could just be that the task is simple enough to where a small number of attention representations is required.

ii This forces the heads to be more independent and cohesive themselves, instead of relying on any one other head being present. Individually, the heads must be useful themselves and therefore must learn more about the context.